Environmental Non-Screening of the Z6 Chronometric Ratio Mode

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Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral; see RECOVERY_PROVENANCE.md.

26. Nonlinear environmental equation

Let $x = a/f_a$. The static spherical equation is

$$\nabla^2 x = -\frac{m_a^2}{6}\sin 6x + \sum_A \frac{\rho_A}{f_a^2} p_A \epsilon \sin x (1-\epsilon\cos x)^{p_A-1}.$$

The benchmark

$$M = 10\,\mathrm{TeV}, \qquad f_a = M_{\mathrm{P}}, \qquad \epsilon = 10^{-6}$$

has

$$m_a = 3.797\times 10^{-21}\,\mathrm{eV}, \qquad \lambda_a = 347.4\,\mathrm{AU}.$$

Homogeneous spinodal versus finite-body conversion

Define the local density parameter

$$t = \frac{\rho p \epsilon}{m_a^2 f_a^2}.$$

The branch connected to $x_\infty = \pi/2$ disappears in an infinite homogeneous medium above

$$t_* = 0.17272340294,$$

corresponding in the benchmark to

$$\rho_{\mathrm{spinodal}} = 46.25\,\mathrm{g\,cm^{-3}}.$$

The solar core exceeds this local threshold. A finite body must also pay gradient energy. For a core of radius R_c, define

$$q(R_c) = \frac{M(<R_c)p\epsilon}{4\pi f_a^2 R_c}.$$

Conversion to the adjacent phase requires at least

$$\boxed{q(R_c) > q_{\mathrm{conv}} = 0.633135.}$$

Density can prefer another phase locally, but finite objects are much too weakly compact to realise it.

Figure 1: Density can prefer another phase locally, but finite objects are much too weakly compact to realise it.

Earth, Sun, and a 5 cm tungsten source fall short by enormous margins.

Source	Largest q/q_{conv}	Surface shift	Screening factor
Earth	1.63×10^{-16}	-1.03×10^{-16}	1.000000
Sun	1.02×10^{-12}	-3.15×10^{-13}	1.000000
5 cm tungsten	3.50×10^{-32}	-2.22×10^{-32}	1.000000

The Sun therefore provides the desired source without screening it. The field's response length is vastly larger than the source, so it reacts to integrated scalar charge rather than tracking a local density minimum. This is a thick-shell/no-shell regime rather than a chameleon thin shell [@Chameleon2008; @FiniteDensity2024].

Finite-size bound

Using $G = (8\pi M_{\mathrm{P}}^2)^{-1}$,

$$q = 2p\epsilon \left(\frac{M_{\mathrm{P}}}{f_a} \right)^2 \Phi,$$

where $\Phi = GM/(Rc^2)$. If

$$f_a \geq M_{\mathrm{P}}, \qquad \epsilon \leq 1, \qquad p \leq \frac{2}{27},$$

then any nonsingular spherical body outside a trapped surface has

$$q < \frac{2}{27} < q_{\mathrm{conv}}.$$

This excludes complete matter-driven conversion for that parameter class, not cosmological domains or more elaborate UV interactions.